

Olive Knot AI

Computer vision and plant research

National second place in Software
TÜBİTAK 2204-A, 2022

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Research question



Can a plant image help identify the visible stage of olive knot disease?

A CNN-based classifier grouped images into four stages. I put the predictions into a mobile research prototype.

Research prototype — not a certified diagnosis.

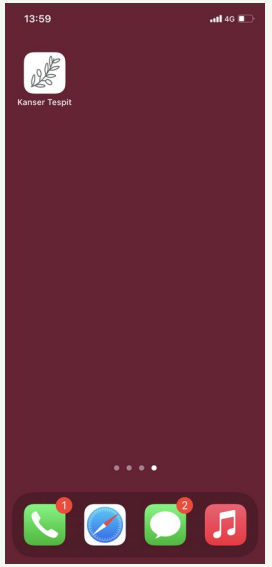
Mobile prototype



98.03% • stage 1

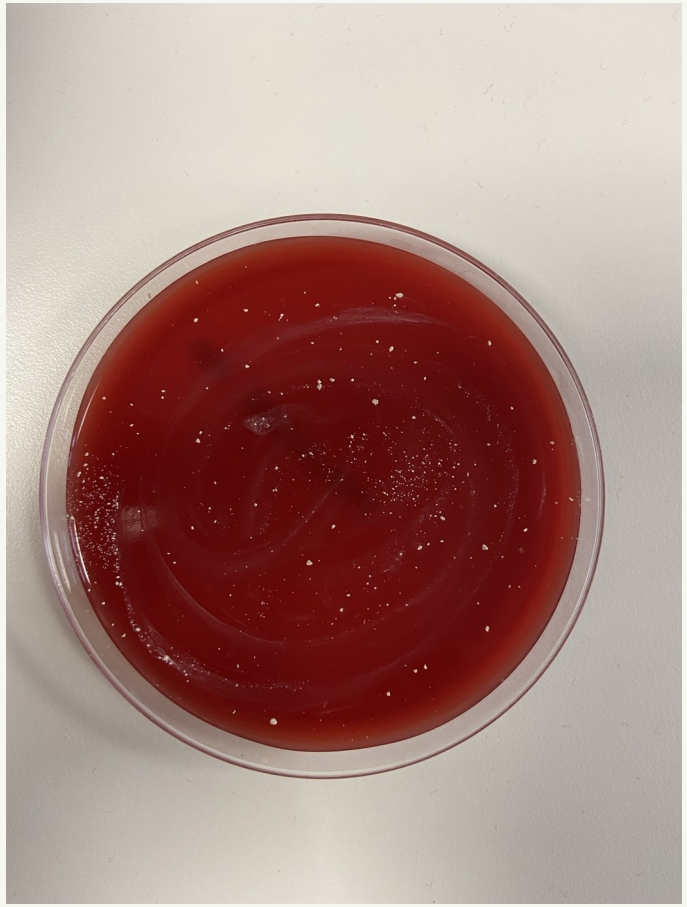


93.27% • stage 4



Confidence on these two images, not an overall accuracy measure.

Plant and laboratory study

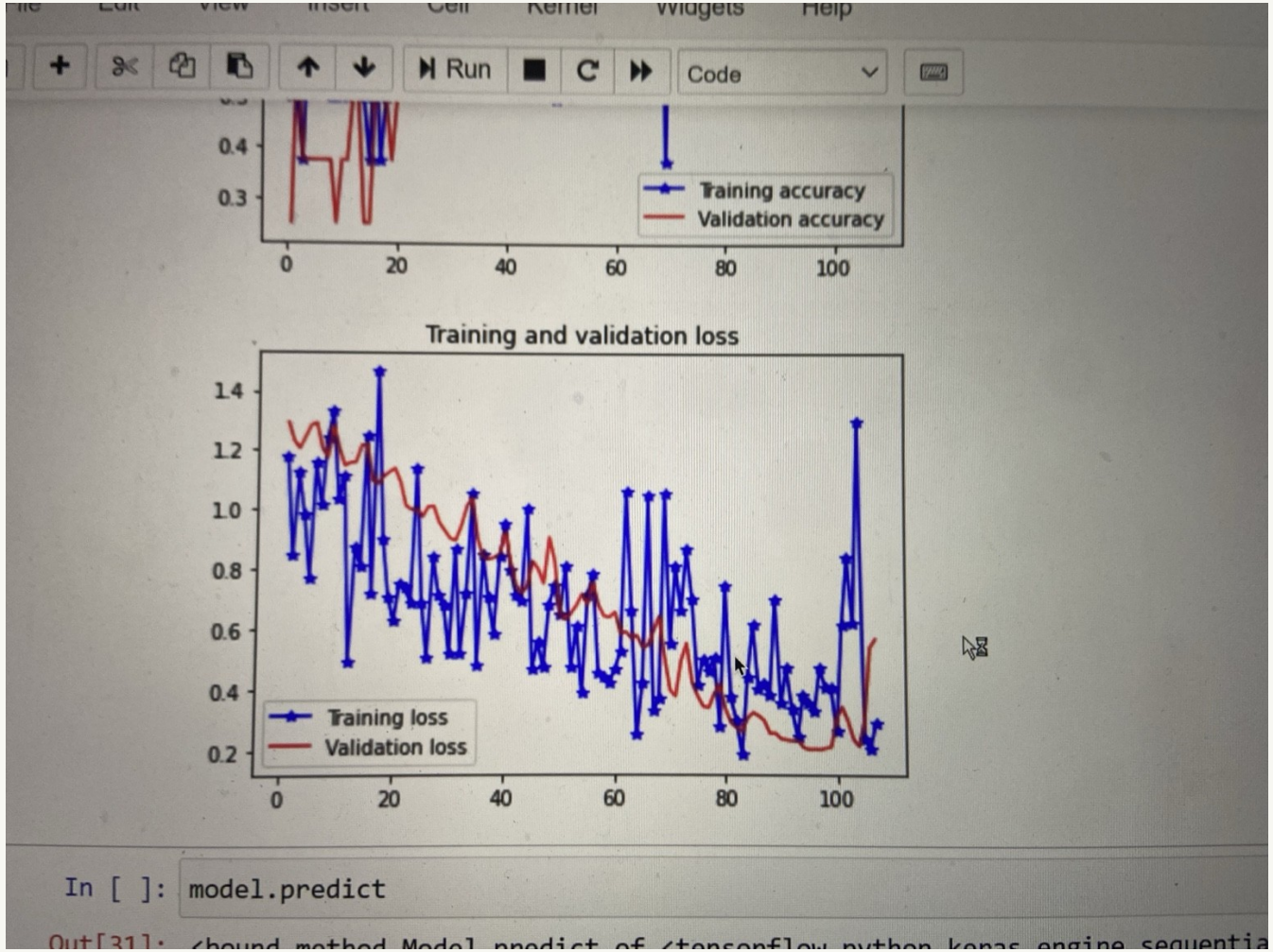


12 olive saplings

Four comparison groups
About four months of observation

The poster reports preliminary observations about disease progression and Bacillus subtilis. More samples and independent replication would be needed for a treatment claim.

Training and evaluation



88.33%

classification accuracy reported in the 2022 poster

Self-reported research result. No independent field validation is claimed.

Regional and national recognition



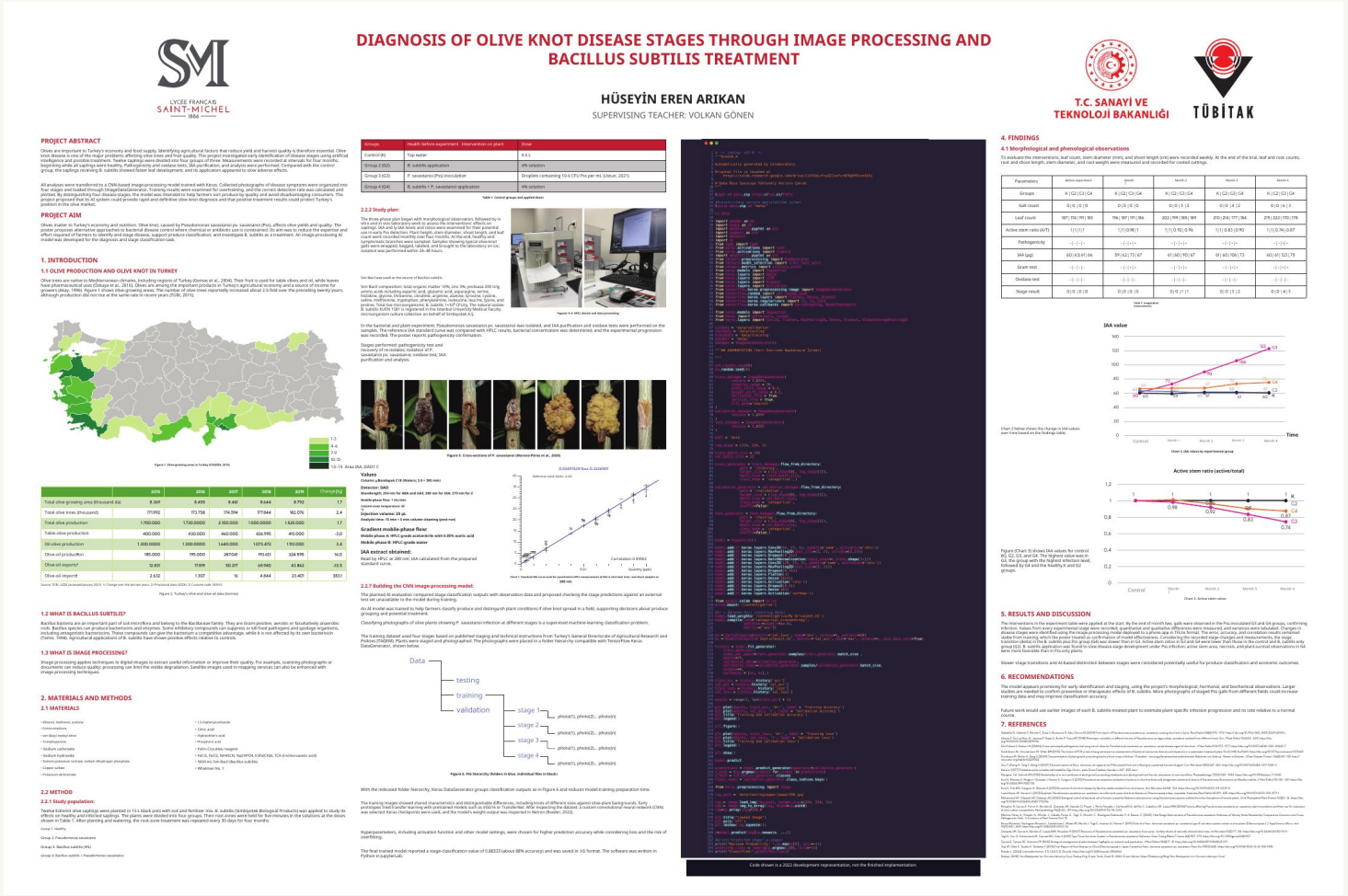
Regional first place

National second place in Software

TÜBİTAK 2204-A • 2022

25,095 project submissions and 37,283 students across all 12 fields.

The research poster in English



DIAGNOSIS OF OLIVE KNOT DISEASE STAGES THROUGH IMAGE PROCESSING AND BACILLUS SUBTILIS TREATMENT

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TÜBİTAK

PROJECT ABSTRACT

Olives are important to Turkey's economy and food supply. Identifying agricultural factors that reduce yield and harvest quality is therefore essential. Olive knot disease is one of the major problems affecting olive trees and fruit quality. This project investigated early identification of disease stages using artificial intelligence and possible treatment. Twelve saplings were divided into four groups of three. Measurements were recorded at intervals for four months, beginning while all saplings were healthy. Pathogenicity and oxidase tests, IAA purification, and analysis were performed. Compared with the control group, the saplings receiving B. subtilis showed faster leaf development, and its application appeared to slow adverse effects.

All analyses were transferred to a CNN-based image-processing model trained with Keras. Collected photographs of disease symptoms were organized into four classes corresponding to the stages of disease progression. Training and validation accuracy and loss values were recorded for each epoch and plotted. By distinguishing four disease stages, the model was intended to help farmers sort produce by quality and avoid disadvantaging consumers. The project proposed that its AI system could provide rapid and definitive olive-knot diagnosis and that positive treatment results could protect Turkey's position in the olive market.

PROJECT AIM

Olives matter to Turkey's economy and nutrition. Olive knot, caused by Pseudomonas savastanoi pv. savastanoi (Psv), affects olive yields and quality. The poster proposes alternative approaches to bacterial disease control when chemical or antibiotic use is constrained. Its aim was to reduce the expertise and effort required of farmers to identify and stage disease, support produce classification, and investigate B. subtilis as a treatment. An image-processing AI model was developed for the diagnosis and stage classification task.

1. INTRODUCTION

1.1 OLIVE PRODUCTION AND OLIVE KNOT IN TURKEY

Olive trees are native to Mediterranean climates, including regions of Turkey (Gemas et al., 2004). Their fruit is used for table olives and oil, while leaves have pharmaceutical uses (Özkaya et al., 2015). Olives are among the important products in Turkey's agricultural economy and a source of income for growers (Akay, 1996). Figure 1 shows olive-growing areas. The number of olive trees reportedly increased about 2.5-fold over the preceding twenty years, although production did not rise at the same rate in recent years (TUİK, 2019).

Figure 1. Olive-growing areas in Turkey (TUGEM, 2016).

	2015	2016	2017	2018	2019	Change (%)
Total olive-growing area (thousand da)	8.369	8.435	8.461	8.644	8.792	1.7
Total olive trees (thousand)	171992	173.756	174.094	177.644	182.076	2.4
Total olive production	1.700.0000	1.730.0000	2.100.0000	1.500.00000	1.525.0000	1.7
Table-olive production	400.0000	430.000	460.0000	426.995	410.0000	-3.0
Oil-olive production	1.300.00000	1.300.00000	1.640.0000	1.073.672	1.150.0000	3.4
Olive-oil production	195.0000	195.7500	287.5581	193.615	224.595	16.0
Olive-oil exports*	12.831	17.899	50.217	69.940	43.862	-33.5
Olive-oil import#	2.632	1.307	16	4.844	23.401	383.1

Source: TUİK, UZEM (accessed January 2021). *) Change over the last two years, #) Provisional data (2019). $) Customs code: 150910.

Figure 2. Turkey's olive and olive-oil data (tonnes).

1.2 WHAT IS BACILLUS SUBTILIS?

Bacillus bacteria are an important part of soil microflora and belong to the Bacillaceae family. They are Gram-positive, aerobic or facultatively anaerobic bacteria. Bacillus species can form endospores, which are resistant to unfavorable conditions. Some organisms in left-hind pathogens and spoilage organisms, including antagonistic bacteriocins. These compounds can give the bacterium a competitive advantage, while it is not affected by its own bacteriocin (Tiwesi, 1998). Agricultural applications of B. subtilis have shown positive effects relatives to control.

1.3 WHAT IS IMAGE PROCESSING?

Image processing applies techniques to digital images to extract useful information or improve their quality. For example, scanning photographs or documents can reduce quality; processing can limit the visible degradation. Satellite images used in mapping services can also be enhanced with image-processing techniques.

2. MATERIALS AND METHODS

2.1 MATERIALS

- Ethanol, methanol, acetone
- Dichloromethane
- Iron (ferric) methylene
- Trimethylamine
- Sodium carbonate
- Sodium hydroxide
- Tetrahydrosodium tetrazinc sodium dihydrogen phosphate
- Copper sulfate
- Potassium dichromate
- 1,5-Diphenylcarbazide
- Citric acid
- Hydrochloric acid
- Phosphoric acid
- Folin-Ciocalteu reagent
- FeCl3, FeCl2, NH4Cl2S, Na2HPO4, KOHxCNSJ, TCA (trichloroacetic acid)
- 5000 mL Ion-Starch Bacillus subtilis)
- Whatman No. 1

2.2 METHOD

2.2.1 Study population:

Twelve 6-month olive saplings were planted in 15 L black pots with soil and fertilizer mix. B. subtilis (Simbiyotek Biological Products) was applied to study its effects on healthy and infected saplings. The plants were divided into four groups. Their root zones were held for five minutes in the solutions at the doses shown in Table 1. After planting and watering, the root-zone treatment was repeated every 30 days for four months.

Group 1. Healthy

Group 2. Pseudomonas savastanoi

Group 3. Bacillus subtilis (PB)

Group 4. Bacillus subtilis + Pseudomonas savastanoi

Groups	Health before experiment - Intervention on plant	Dose
Control (K)	Tap water	0.5 L
Group 2 (G2)	B. subtilis application	4% solution
Group 3 (G3)	P. savastanoi (Psv) inoculation	Droplets containing 10-6 CFU Psv per mL (Unlur, 2021)
Group 4 (G4)	B. subtilis + P. savastanoi application	4% solution

Table 1. Control groups and applied doses

2.2.2 Study plan:

The three-phase plan began with morphological observation, followed by in vitro and in vivo laboratory work to assess the interventions' effects on plants. Table 2 lists the tests and measurements carried out and the tools in use in early Psv detection. Plant height, stem diameter, shoot length, and leaf count were recorded monthly over four months. At the end, healthy and diseased tissues were collected from the plants. Pathogenicity tests (plant stems were wrapped, bagged, labeled, and brought to the laboratory on ice; isolation was performed within 24–48 hours.

Figure 4. SPSS device and discriminating

Sem Bact have used as the source of Bacillus subtilis.

Sim Bact composition: total organic matter 10%; zinc 3%; potassium 200 U/g; phosphorus 200 U/g; nitrogen 3%; magnesium 1%; iron 1%; citric acid; tryptophan; histidine; glycine; thiamine; citrulline; arginine; alanine; tyrosine; cystine; valine; methionine; tryptophan; phenylalanine; isoleucine; leucine; lysine; and proline. Sim Bact is registered in the Istanbul University Medical Faculty microorganism culture collection on behalf of Simbiyotek A.Ş.

In the bacterial and plant experiment, Pseudomonas savastanoi pv. savastanoi was isolated, and IAA purification and oxidase tests were performed on the bacteria. Then, the effects of B. subtilis on the plants were observed. According to HPLC results, bacterial concentration was determined, and the experimental progression was recorded. The poster reports pathogenicity confirmation.

Stages performed: pathogenicity test and recovery of re-isolates; isolation of P. savastanoi pv. savastanoi; oxidase test; IAA purification and analysis.

Figure 5. Cross-sections of P. savastanoi (Moreno-Pérez et al., 2020)

Figure 6. Standard IAA curve used for quantitative HPLC measurement of IAA in olive knot, host, and shoot samples at 280 nm.

2.2.7 Building the CNN image-processing model:

The planned AI evaluation compared stage-classification outputs with observation data and proposed checking the stage predictions against an external test set unavailable to the model during training.

An AI model was trained to help farmers classify produce and distinguish plant conditions if olive knot spread in a field, supporting decisions about produce grouping and potential treatment.

Classifying photographs of olive plants showing P. savastanoi infection at different stages is a supervised machine-learning classification problem.

The training dataset used four stages based on published staging and technical instructions from Turkey's General Directorate of Agricultural Research and Policies (TAGEM). Plants were staged and photographed. The photographs were placed in a folder hierarchy compatible with Tensorflow Keras DataGenerator, shown below.

Figure 8. File hierarchy (folders in blue, individual files in black)

With the indicated folder hierarchy, Keras DataGenerator groups classification outputs as in Figure 6 and reduces model-training preparation time.

The training images showed shared characteristics and distinguishable differences, including kinds of different sizes against olive plant backgrounds. Early convolutional layers of the model were expected to identify such features. Figure 7 shows the model's architecture. A custom convolutional neural network (CNN) was selected. Keras checkpoints were used, and the model's weight output was inspected in Netron (Roeder, 2022).

Hyperparameters, including activation function and other model settings, were chosen for higher prediction accuracy while considering loss and the risk of overfitting.

The final trained model reported a stage-classification value of 0.88333 (about 88% accuracy) and was saved in .h5 format. The software was written in Python in Jupyterlab.

Code shown is a 2022 development representation, not the finished implementation.

4. FINDINGS

4.1 Morphological and phenological observations

To evaluate the interventions, leaf count, stem diameter (mm), and shoot length (cm) were recorded weekly. At the end of the trial, leaf and root counts, root and shoot length, stem diameter, and root weight were measured and recorded for rooted cuttings.

Parameters	Before experiment	Month 1	Month 2	Month 3	Month 4
Groups	K | G2 | G3 | G4	K | G2 | G3 | G4	K | G2 | G3 | G4	K | G2 | G3 | G4	K | G2 | G3 | G4
Gall count	0 | 0 | 0 | 0	0 | 0 | 0 | 0	0 | 0 | 3 | 2	0 | 0 | 4 | 2	0 | 0 | 6 | 3
Leaf count	187 | 176 | 191 | 185	196 | 187 | 191 | 186	202 | 199 | 185 | 189	210 | 214 | 177 | 194	209 | 222 | 191 | 178
Active stem ratio (A/T)	1 | 1 | 1 | 1	1 | 1 | 0.92 | 0.96	1 | 1 | 0.83 | 0.90	1 | 1 | 0.83 | 0.87	1 | 1 | 0.74 | 0.87
Pathogenicity	- | - | - | -	- | - | - | -	- | - | - | -	- | - | - | -	- | - | - | -
IAA (µg)	6.0 | 6.2 | 6.1 | 6.6	59 | 6.2 | 7.5 | 6.7	6.1 | 6.0 | 9.0 | 6.7	6.1 | 6.0 | 7.6 | 7.5	6.0 | 6.1 | 12.5 | 7.5
Gram test	- | - | - | -	- | - | - | -	- | - | - | -	- | - | - | -	- | - | - | -
Oxidase test	- | - | - | -	- | - | - | -	- | - | - | -	- | - | - | -	- | - | - | -
Stage result	0 | 0 | 0 | 0	0 | 0 | 0 | 0	0 | 0 | 1 | 1	0 | 0 | 3 | 2	0 | 0 | 4 | 3

Table 2. Longitudinal measurements

Chart 2 below shows the change in IAA values over time based on the finding table.

Chart 2. IAA values by experimental group

Figure (Chart 3) shows IAA values for control (K) G2, G3, and G4. The highest value was in G3, the group with the highest infection level, followed by G4 and the healthy K and G2 groups.

Chart 3. Active stem ratios

5. RESULTS AND DISCUSSION

The interventions in the experiment table were applied at the start. By the end of month two, galls were observed in the Psv-inoculated G3 and G4 groups, confirming infection. Pathogenicity tests and oxidase tests were performed on the bacteria. Then, the effects of B. subtilis on the plants were observed. According to HPLC results, disease stages were identified using the image-processing model deployed in a phone app in TFLite format. The error, accuracy, and correlation results remained stable from training, which the poster treated as confirmation of model effectiveness. Considering the recorded stage changes and measurements, the stage transition differed in the B. subtilis plus Psv group (G4) was slower than in G3. Active-stem ratios in G2 and G4 were lower than those in the control and B. subtilis only group (G2). B. subtilis application was found to slow disease-stage development under Psv infection; active stem area, necrosis, and plant-survival observations in G4 were more favorable than in Psv-only plants.

Slower stage transitions and AI-based distinction between stages were considered potentially useful for produce classification and economic outcomes.

6. RECOMMENDATIONS

The model appears promising for early identification and staging, using the project's morphological, hormonal, and biochemical observations. Larger datasets could improve model performance and reduce the negative effects of B. subtilis. More photographs of stages of Psv galls from different fields could increase training data and may improve classification accuracy.

Future work would use earlier images of each B. subtilis-treated plant to estimate plant-specific infection progression and its rate relative to a normal course.

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Demonstrating the work



The project was built, presented and awarded.

The finished code, model weights and training data remain private. The poster and research record are open to inspect.